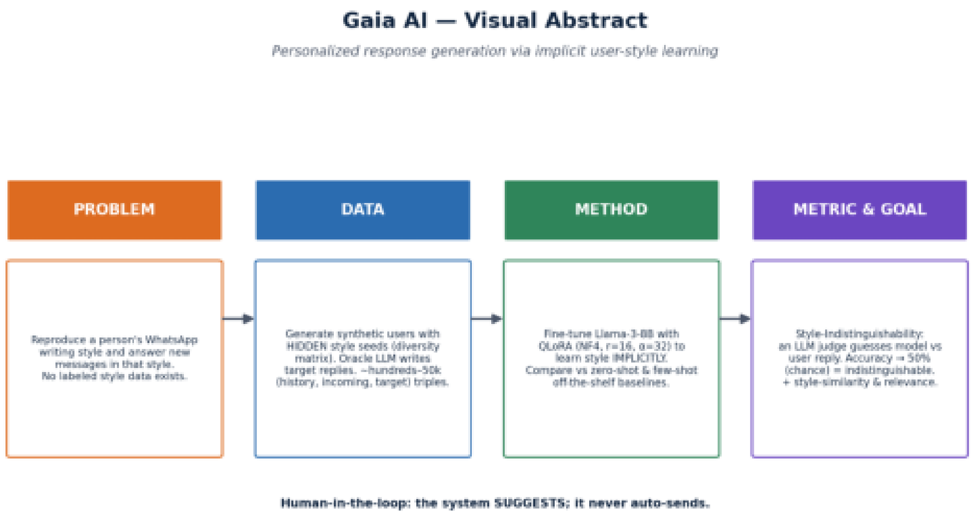
Gaia AI — Final Report

Personalized response generation via implicit user-style learning · Adir · LLM course (HIT)

1 · Project Definition (refined)

- Problem: given a user's chat history + a new message, generate a reply in the user's IMPLICIT style.
- Models: fine-tuned Llama-3-8B (QLoRA) vs off-the-shelf zero-/few-shot; oracle = upper bound.
- Data: synthetic, persona-conditioned, bilingual; per-user splits.
- Metrics: Style-Indistinguishability (headline), style-similarity, relevance.

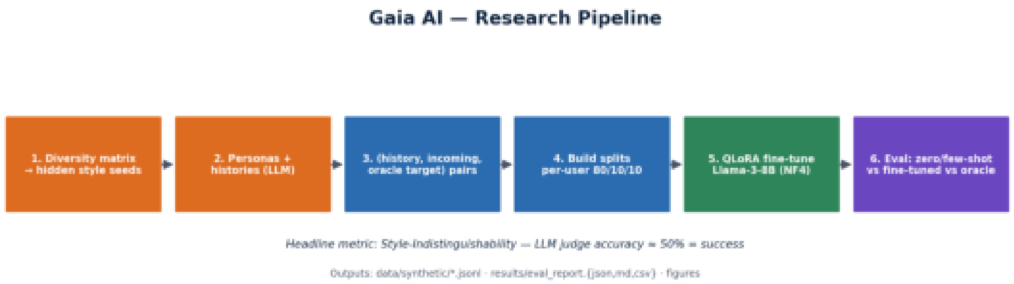


2 · Achievements & Novelty

- Built an end-to-end pipeline: diversity matrix → synthetic data → fine-tune → indistinguishability eval.
- Generated a bilingual persona-conditioned dataset with hidden style seeds (no style labels leaked).
- Produced real baseline results and a reproducible eval (JSON/MD/CSV + figures).
- Novelty: implicit style learning + indistinguishability metric — distinct from retrieval-based personalization (LaMP) and explicit personas (PersonaChat).

3 · Methodology

- Synthetic generation: LLM role-play with hidden seeds; oracle writes targets; robust JSON parsing.
- Fine-tuning: QLoRA (4-bit NF4, r=16, α=32, target q/k/v/o) on Llama-3-8B-Instruct (cloud GPU).
- Evaluation effort for quality:
  - Per-user splits; order-randomized judge; judge model separated from generator.
  - Multilingual embedder for he+en style similarity; relevance guards content.
  - Rate-limit-aware generation (backoff, serialized) for reproducibility on free tiers.



4 · Results

- Run eval.run_all + ml/eval/plots.py to populate real numbers and figures.

Model	Indist.	95% CI	StyleSim	Relev.
zero_shot	pending	pending	pending	
few_shot	pending	pending	pending	
fine_tuned	GPU run	—	—	

Indistinguishability ≈ 50% = success. Figures in visuals/results/.

5 · Conclusion & Future Work

- The framework works end-to-end and yields real baseline numbers; fine-tuned arm is one GPU run away.
- What I'd change: larger dataset (paid LLM tier), a stronger separate judge model, more error analysis.
- Future: real opt-in data generalization test; RLHF from user edits; multi-personality switching.
- Human-in-the-loop throughout — the system suggests; the user approves.